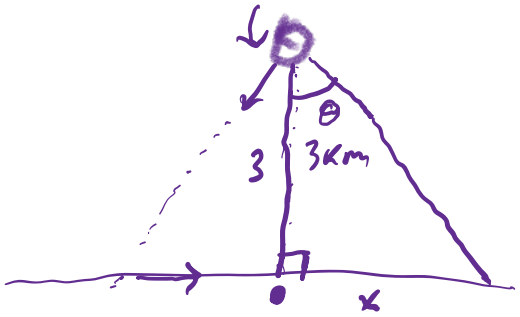
Problem 14. TIER 4 — ANGLE GIVEN, LENGTH WANTED

A lighthouse stands 3 km from a straight shoreline, and its light rotates at 4 revolutions per minute. How fast is the spot of light moving along the shore when it is 1 km from the point on the shore nearest the lighthouse?

(1)



(2) $\frac{d\theta}{dt} = 4\frac{\text{rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} = 8\pi \frac{\text{rad}}{\text{min}}$

$\frac{dx}{dt} = ?$ $\quad x = 1$

(3) $\tan\theta = \frac{x}{3} \Rightarrow x = 3\tan\theta$

(4) $\frac{d}{dt}x = \frac{d}{dt}3\tan\theta$

$\frac{dx}{dt} = 3\sec^2\theta \frac{d\theta}{dt}$

(5)



$\sqrt{3^2+1^2} = \sqrt{10}$

$\sec\theta = \frac{\text{hyp}}{\text{adj}}$

$\sec\theta = \frac{\sqrt{10}}{3} \rightarrow \sec^2\theta = \frac{10}{9}$

$\frac{dx}{dt} = 3 \cdot \frac{10}{9} 8\pi$

$\frac{dx}{dt} = \frac{80\pi}{3} \frac{\text{km}}{\text{min}}$